20 A 2200 μF capacitor is initially charged to a potential difference of 9.0 V. The initial amount of energy stored in the capacitor is E_o. The capacitor then discharges through a 4.7 kΩ resistor.

How long will it take for the energy stored in the capacitor to fall to $\frac{E_o}{8}$?

A 11 s **B** 16 s **C** 22 s **D** 31 s